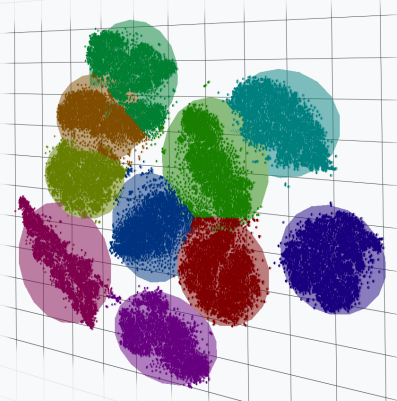
A Framework for Interactive Exploration of Clusters in Massive Data Using 3D Scatter Plots and WebGL

Wagner, Limberger, Scheibel, Trapp, Döllner
Hasso Plattner Institute, University of Potsdam

Web3D 2020



Huge Datasets

Increasingly digital world results in more and more data, overwhelming human comprehension

Huge Datasets

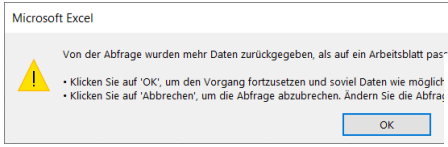
Increasingly digital world results in more and more data, overwhelming human comprehension

Example dataset: Hathi Trust library catalog
9 columns, 5M rows, >600 MB .tsv [1]

[1] Data source: creatingdata.us/data/scatter/hathi/tiles/

First Approach: Use Excel

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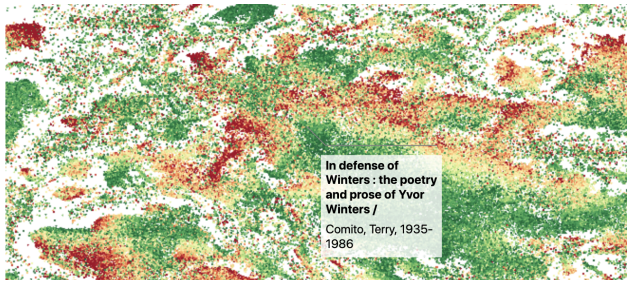
Worksheet and workbook specifications and limits

Feature	Maximum limit
Open workbooks	Limited by available memory and system resources
Total number of rows and columns on a worksheet	1,048,576 rows by 16,384 columns

5M rows is too much data
for Excel to visualize

Second Approach: Implement Viewer for a Specific Dataset

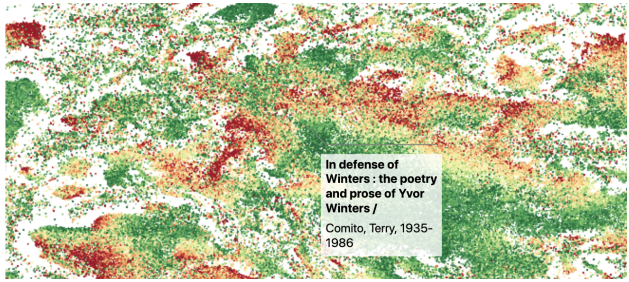
Colin Eberhardt:
“Rendering One Million Datapoints with D3 and WebGL” [2,3]



[2] Blog article: blog.scottlogic.com/2020/05/01/rendering-one-million-points-with-d3.html
[3] Demo: colineberhardt.github.io/d3fc-webgl-hathi-explorer/

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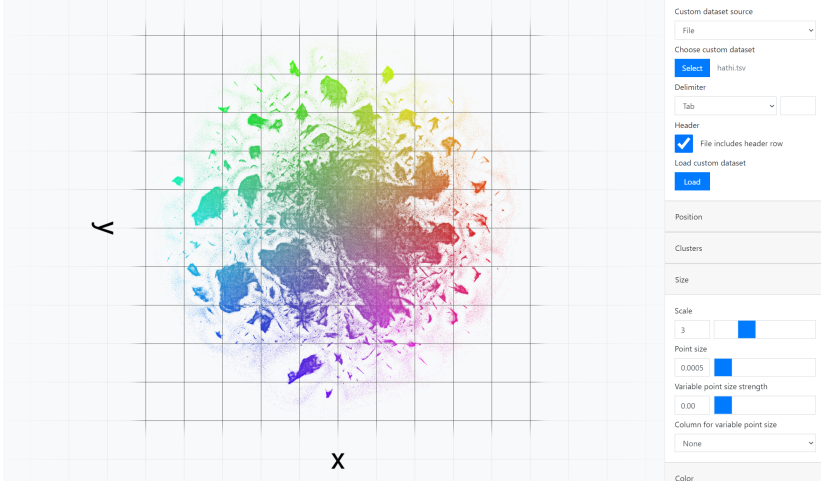
Works well, but optimized for specific dataset

[2] Blog article: blog.scottlogic.com/2020/05/01/rendering-one-million-points-with-d3.html
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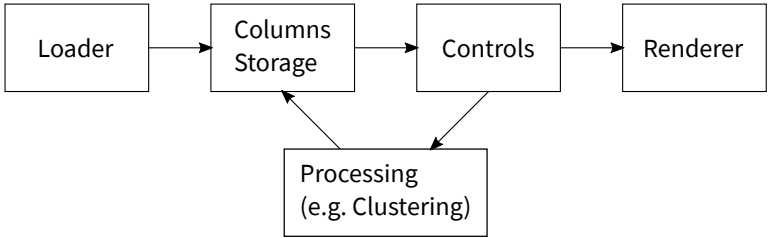
Viewer Requirements

Huge datasets	Support datasets containing millions of datapoints
Dataset agnostic	Allow arbitrary mapping of columns to visual variables
Ease of access	Web-based implementation for cross-platform support without installation

Demo


Y
X
Custom dataset source
File
Choose custom dataset
Select
hathi.tsv
Delimiter
Tab
Header
File includes header row
Load custom dataset
Load
Position
Clusters
Size
Scale
3
Point size
0.0005
Variable point size strength
0.00
Column for variable point size
None
Color

Approach



Central storage of columns

- fed by modular loader (.csv, .tsv, ...)
- provides columns to be mapped to visual variables
- columns can also be used for processing steps

Approach

When configuring an output, only compatible columns are offered for selection

Visual variables:

- {name: "x-coord", type: "number"}
- {name: "y-coord", type: "number"}
- {name: "z-coord", type: "number"}
- {name: "size", type: "number"}
- {name: "color", type: "color"}

Available columns:

- {name: "year", type: "number"}
- {name: "country", type: "text"}
- {name: "ranking", type: "number"}
- {name: "score", type: "number"}
- {name: "jersey", type: "color"}

Approach

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Technology Stack



Rendering API:
WebGL 2



Framework:
webgl-operate



Language:
TypeScript

Optimized data handling

Data is loaded and processed using Web Workers

- allows for parallel computing
- keep UI interactive by offloading work from main thread

Web Workers are separate JS runtimes

- data has to be copied between workers
- solution: SharedArrayBuffers map the same memory location to multiple runtimes, improving speed and memory consumption

Rendering Approach

Rendering of millions of datapoints

- instanced rendering of a quad [4]
- mapped columns are passed as per-instance attributes

Separation of rendering into passes

- points, clusters, grid, labels are rendered by separate classes
- modularity allows for easy adjustments and extension

[4] developer.mozilla.org/en-US/docs/Web/API/WebGL2RenderingContext/drawArraysInstanced

Performance Measurements

Test System in 1920×1080px	**Number of Points** 1×10^6	5×10^6	1×10^7	5×10^7	1×10^8
Intel UHD 630	26.8	18.6	10.8	2.6	1.4
Nvidia GTX 1650	196.6	65.2	38.0	8.9	4.6
Nvidia GTX 1080	474.6	154.3	81.2	17.6	9.0

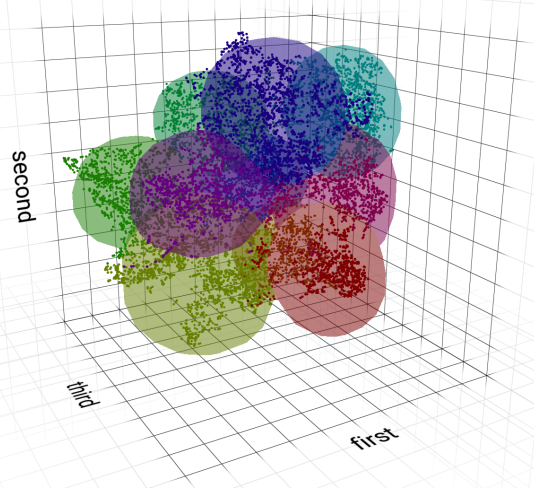
Table: Frames per second for different amounts of data measured on a low, mid, and high end GPU in Chrome 84.

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Try it out: varg.dev/web3d2020